

- Hide menu
- Week 2: Image Formation
- Reading: Week 2 Lecture Handout

30 min
- Ungraded Plugin: 2.1

Overview of Image Formation

3 min
- Practice Quiz: 2.1

Overview of Image Formation Self-check Quiz

Started
- Ungraded Plugin: 2.2

Pinhole and Perspective Projection

21 min
- Practice Quiz: 2.2

Pinhole and Perspective Projection Self-check Quiz

Started
- Reading: 2.3 Video

Correction

10 min
- Ungraded Plugin: 2.3

Image Formation using Lenses

12 min
- Practice Quiz: 2.3

Image Formation using Lenses Self-check Quiz

Started
- Ungraded Plugin: 2.4

Depth of Field

15 min
- Practice Quiz: 2.4

Depth of Field Self-check Quiz

Started
- Ungraded Plugin: 2.5

Lens Related Issues

8 min
- Practice Quiz: 2.5

Lens Related Issues Self-check Quiz

5 min
- Ungraded Plugin: 2.6

Wide Angle Cameras

24 min
- Ungraded Plugin: 2.7

Animal Eyes

16 min
- Discussion Prompt: Week 2

Intrusive Detection System

10 min
- Week 2 Quiz

Congratulations! You passed!

Grade received 100% To pass 50% or higher

Go to next item

2.4 Depth of Field Self-check Quiz

1.

If a point in 3D space produces a blur circle with radius greater than pixel size, then it is in focus.

1 / 1 point

True

False

Ready to review what you've learned before taking the quiz? I'm here to help.

Help me practice

Let's chat

Correct

A point in 3D space produces a blur circle with radius greater than pixel size, then it is in focus.

Try again

2.

An object at a distance 20 cm from the lens produces a blur circle with a radius 1 pixel size. If you move these objects any further the blur circle might be greater than the pixel size. Calculate the depth of field (DoF) for the system.

1 / 1 point

80mm

0.8m

40mm

60mm

Like

Dislike

Report an issue

Correct

Use the following formula: 60mm - 20 mm = 40mm

3.

An object (Object R) at 50mm from lens produces image on an image plane. There are two more objects - Object 1 is at 5mm away from the object R and object 2 is 5mm in front of the object R. Assuming they all produce blur circle within a pixel size, what can you say about the amount of blur produced by both objects.

1 / 1 point

Both objects produce same amount of blur

Object 2 produced more blur than object 1

Object 1 produces more blur than object 2

Can not determine from the information given.

Correct

The farther you move from the plane of focus, the degree of defocus increases but the magnitude of increase is not the same on both sides. The closer you move towards the lens the defocus increases rapidly compared to the farther you move.